



CSE 250L

Circuits and Electronics Lab

Marks Distribution:

Type	Assessment	Weight	# of Assessments to be Assigned	# of Assessments to be Graded
Lab (25%)	Attendance	2%	6 experiments	90% or more to attend the final exam
	Lab Performance	5%	6 experiments	Best 6 (=n-1), No make-up lab
	Lab Report (Hardware)	6%	2<	Best: 4% 2nd Best: 2% Individual
	Lab Test	12%	2	Best one: 8% Individual
	Total	25%		

Lab Experiments & Probable Timeline:

No.	Experiment Name	Type	Week/Experiment No.	Related CO (if any)
1	Introduction to Laboratory Instruments (Part 1)	Hardware	Week 2 / Experiment 0	CO4
2	Introduction to Series and Parallel Circuits.	Hardware	Week 2 / Experiment 1	CO4
3	Verification of KVL and KCL.	Hardware	Week 3 / Experiment 2	CO4
4	Verification of the Superposition Principle.	Hardware	Week 3 / Experiment 3	CO4
5	Open practice for lab tests	Hardware	Week 4 / Practice	
6	Lab Test 1	Hardware	Week 5 / Exam	CO6
Midterm Week				
7	Study of I-V Characteristics and Circuit Equivalence.	Hardware	Week 8 / Experiment 4	CO4
8	Verification of Thevenin's Theorem and Maximum Power Transfer Theorem.	Hardware	Week 9 / Experiment 6	CO4
9	Study of Transient Behaviour of RC Circuit.	Hardware	Week 10 / Experiment 7	CO4
10	Open practice for lab tests	Hardware	Week 11 / Practice	
11	Lab test 2	Hardware	Week 12 / Exam	CO6
Final Week				

Lab Calendar (*subject to change*):

(will be updated soon)

Group Formation:

- There will be 6 groups in each lab section. The groups will be formed before starting the first lab experiment. The formed groups will be maintained throughout the semester.
- The lab faculty members hold the power to form/change/take any decision regarding the lab groups.

Attendance and Class Performance:

- Class performance marks will be calculated by adding the class performance marks from all the lab days (10 marks each).
- A student's attendance will be counted as "Late Attendance" if a student comes 15 minutes after the scheduled lab time. Every "late attendance" will be considered as "half attendance", and the class performance will also be counted as half for that day.
- If a student misses an experiment due to a valid reason, they must make up for it in another section, subject to approval from one of the lab faculty members of that section. To do so, the student must first obtain permission from their own section's lab faculty. Then, they must email the faculty members of the section they wish to attend for the makeup, keeping their own section's faculty member in CC. Failure to follow this process will result in no attendance or lab performance marks for that experiment under any circumstances.
- The student will be barred from attending the finals if a student fails to attend more than one lab session or accumulate more than 2 late attendances throughout the semester.

Lab Reports and Assignments:

- Lab reports and lab assignments are individual.
- One individual from each group will have to submit at least 2 lab reports on two *hardware* experiments (one from experiments 1 to 3 and another from experiments 4, 6, and 7). The submissions for the hardware lab reports will be in a rotating manner. For each experiment, there can be a maximum of two or a minimum of one report from each group for each experiment.
- A student must submit both the hard copies and the soft copies of the hardware lab reports. The **deadline** for the softcopy submission is 11:59 PM on the day before the lab day of the following week. Faculty will provide a Google Form link for the report softcopy submission. In case a lab experiment on a particular topic is conducted before the topic is covered in theory classes; the deadline for report submission will be adjusted accordingly by the lab faculty (*with the exception of Experiment 3*).
- For the hardware labs, at least one person from each group must submit a report on every experiment. If not, marks will be deducted from the person who was responsible for the submission.
- By the time all the experiments are completed, each student should have at least two lab reports submitted. In case a student submits more than two hardware reports, the best 2 will be counted.
- The report must be prepared in the provided lab sheet. Students must answer all the questions and show the necessary calculations. The Discussion and the Data Table sections should also be completed properly. Extra pages can be used if necessary.
- N.B.: The hard copy of the lab sheet will be provided by the faculty in daily lab sessions (one copy for each group). The softcopy will be provided in [Discord](#).

Lab Tests:

- Lab tests will be based on the hardware experiments only (Experiments 1 - 3, 4, 6, & 7).
- Two hardware lab tests will be held after performing the hardware experiments (Experiments 1 - 3, 4, 6, & 7). In the case of two lab tests, the first one will be held before midterms, and Lab Test 2 will be held before finals.
- The tests are individual.
- Students may be asked to implement a circuit comprising series-parallel combinations of circuit components and measure voltage, current, power, etc., using a multimeter and plot waveforms using the oscilloscope. The understanding and interpretation skills of the students will be tested through a viva during the lab test.

- * *All the policies are subject to change depending on the decisions of the faculty members.*
- * *Any kind of plagiarism will result in severe punishment.*